

1 **FACT STATEMENT OF LOGAN COBB, M.D.**

2
3 Greetings! I am Logan Cobb, the chief medical examiner for the City of Texoma for the last eight
4 years. I am also an expert in forensic pathology and forensic toxicology. I'm 56 years old and
5 happily married. We have three wonderful children. They're all grown up now, so we're enjoying
6 our new status as empty nesters with our two pet corgis that we adopted last year. Outside of work,
7 I enjoy fishing and golf. I can never get lucky with the former and am no good at the latter, but
8 maybe one day. I graduated from Harvard University in 1995 with a double major in biology and
9 chemistry. I originally thought I'd work at a big pharmaceutical company as a chemist, but my
10 mother's dying wish was that I go to medical school. I graduated from Johns Hopkins four years
11 later, earning my M.D., then spent my residency in pathology. After that, I worked for two years
12 at the Mayo Clinic where I specialized in medical and forensic toxicology on top of forensic
13 pathology. For those who are unfamiliar, a forensic pathologist is a medical doctor who specializes
14 in and investigates sudden, unexpected, suspicious, or violent deaths. By nuanced contrast, a
15 forensic toxicologist is a medical doctor specializing in identifying and quantifying drugs, poisons,
16 intoxicants, and toxins in a person or biological sample.

17
18 The Mayo Clinic was where I got my big break. A friend of mine at the Texoma Bureau of
19 Investigation (TBI) asked me to consult on a cold case – an unsolved poisoning of a famous
20 Texoma actress, Delilah James. She was the first woman from Texoma to win an Oscar for Best
21 Actress. During the initial investigation of the murder, the police collected samples from Ms.
22 James's blood because they believed a rare poison had been used to kill her. But they got zero
23 leads for nearly two decades. I was able to test the blood using a new method; I found trace levels
24 of bacteria in the blood that originate only from a specific type of soil found in the mountains of
25 South America. And guess who had just come back from a hiking trip in Machu Picchu days
26 before the murder – none other than Ms. James's nanny. Turns out, Ms. James had fired the nanny
27 the day she died, clearly motive for murder.

28
29 After I testified at the trial, the case got major international attention in the field of toxicology,
30 which got me published everywhere. My career took off. I consulted for all the major three-letter
31 agencies: TBI, FBI, DEA, you name it. But after all the whirlwind travel and publicity, I decided
32 to stop consulting and settle down in Texoma to spend more time with my family and kids. I was
33 offered the chief medical examiner job and accepted on the spot. In the eight years since, I have
34 performed hundreds of autopsies and regularly consult with local law enforcement agencies,
35 including the Texoma Police Department (TPD), when they have complicated cases that require
36 my toxicological expertise.

37
38 I personally performed the autopsy of Julien Younger, the victim, and authored the autopsy report,
39 which is Exhibit 4. I was on vacation when I received a call from the President of the Texoma
40 State High School Sports Association on November 23, 2025. She was clearly distraught and told
41 me that a student had died after the Texoma State High School Swimming championships. From
42 what I learned, the student was rushed to the hospital after he had collapsed in the locker room –
43 and despite heroic efforts by the medical staff, they were unable to save him. I flew back home a
44 few hours after that call and took charge of the autopsy. I didn't have to do that – a colleague
45 would have covered me while I was on vacation – but I just felt led to do it.

1 The autopsy took place the morning of November 24. Younger was a well-developed male, 18
2 years old, with an athletic build. The body was cool to the touch and rigor mortis had fully
3 developed. The extremities and major body parts were symmetrical and unremarkable. In fact, the
4 entire external examination of Younger's body was unremarkable – but things got interesting when
5 I started the internal examination. Younger's throat and nasal cavities showed signs of extensive
6 lesions with traces of vomit still in the throat. I opened the body with a "Y" shaped incision and
7 immediately noticed two things: the distinct smell of rotten eggs, and extensive internal bleeding
8 that occurred post-mortem. Younger's heart showed definite signs of stenosis in the mid left
9 anterior descending coronary artery and mid right coronary artery. Stenosis is the abnormal
10 narrowing of a body passage, usually through age-related calcium buildup or congenital defects.
11 I saw no congenital defects in Younger's heart and learned there was no family history of heart
12 disease. There was evidence of extensive ulceration and perforations in Younger's esophagus and
13 stomach, the left lung contained a substantial amount of fluid, and numerous organs showed signs
14 of hypoxia, or a lack of oxygen. Based on the totality of the internal damage to Younger's body, I
15 ordered toxicological testing of Younger's blood and tissues since I suspected a toxin had
16 contributed to the death. The results of that testing confirmed my suspicions – high levels of
17 Clartaxin in Younger's blood, one of the most lethal synthetic poisons on the planet. We're talking
18 small-dose-causes-death level toxicity. I determined the manner of death to be homicide with
19 poisoning as the cause of death.

20
21 As for how Clartaxin affects a person, the toxin essentially attacks the body in two ways
22 simultaneously. First, Clartaxin stops the nervous system's ability to communicate with other
23 organs. Normally, when your brain wants a muscle to move, it sends a chemical messenger to the
24 muscle cell; to stop the movement, an enzyme destroys the chemical messenger to allow the muscle
25 to relax. A poison like Clartaxin binds to that enzyme and destroys it, which prevents the enzyme
26 from destroying the chemical messenger. Nerves continuously fire, forcing organs and muscles
27 into a state of endless, uncontrollable overstimulation. The second attack comes from Clartaxin
28 binding to cell surfaces and destroying the ribosomes in the cells that build proteins. These cells
29 quickly die and eventually cause organ failure through acute cellular hypoxia. All of this can occur
30 in approximately 12 to 24 hours after initial exposure.

31
32 The nature of Clartaxin was fully consistent with my observations during Younger's autopsy as
33 well as the reports from first responders, who reported Younger to be spasming with erratic and
34 racing heartbeat and pulse. Notably, there was a relatively high level of internal damage to the
35 victim's head and neck area. This led me to suspect the poison had entered the body somewhere
36 in the head or neck area as the first exposure. I examined the items taken from the victim's locker
37 and was immediately drawn to the black swim goggles. I took swabs of the squishy plastic part
38 on the inside of the goggles that form the seal around the eyes, and I confirmed traces of Clartaxin
39 there. You can see that in clear detail in Exhibit 5, which I took during the testing process. There
40 was no question that was the poison used to kill Younger.

41
42 After I finalized my autopsy report, I spoke at length with Detective Reyne regarding how a killer
43 would make Clartaxin since it's not a naturally occurring toxin. Anyone could theoretically create
44 Clartaxin if they had the right materials, equipment, and a sufficient level of chemistry knowledge.
45 While there is some public information about Clartaxin and its effects, the recipe or formula for
46 how to make it is not something you could just find on the internet or in a book. This is, of course,

1 because of the poison's lethality. There are all kinds of things you can find on the dark side of the
2 internet if you know where to look or who to message, although it's my understanding that the
3 police and feds are constantly monitoring people's activity on the dark web to prevent them from
4 getting their hands on a poison that could be used as a biological weapon. While I am one of the
5 few people who know how to create Clartaxin, I know there are things out there on the dark web
6 and other sketchy places where you can find information like how to make Clartaxin sadly.

7
8 I explained to Detective Reyne what "ingredients" would be needed to make Clartaxin through a
9 chemical extraction process. In short, to make Clartaxin, you need to extract chemicals found in
10 three household items: herbicide, deodorant, and detergent. The problem is you need to apply three
11 vastly different extraction methods to get specific compounds from each item in order to mix them
12 to create Clartaxin. And while you can find the necessary chemicals in common detergents and
13 deodorants, Texoma has banned the specific type of herbicide that you would need; it's only
14 available in France now. The extraction process would take several hours because you'd have to
15 do them one at a time; but once done, you have to mix them together within about twelve hours or
16 the process will not work – it will not produce Clartaxin or anything lethal at all.

17
18 Assuming someone had successfully extracted the chemicals from all three "ingredient" sources,
19 they would still have to combine those chemicals relatively quickly. The mixing process gives off
20 a strong sulfuric aroma, just like the smell of rotten eggs, at multiple points during the process.
21 Sulfur is an extremely recognizable smell, and the aroma created during the mixing process has
22 dense molecular clusters, which means the smell dissipates very slowly. This mixing process is
23 also extremely dangerous because any skin exposure to the Clartaxin would be almost certain
24 death. Protective gear would be essential. While the fumes given off during the mixing process
25 are not as dangerous as direct skin contact, breathing the fumes for any prolonged period would
26 cause significant respiratory problems unless the experiment was in a large or well-ventilated
27 space. Once the three chemicals have been combined, the result is a liquid that must be slowly
28 evaporated using specific equipment, which leaves behind a white paste – Clartaxin.

29
30 I identified the brands of herbicide, deodorant, and detergent that contain the necessary chemicals
31 to make the toxin. I also explained the equipment and other materials someone would need in the
32 process. You'd need much more than some basic laboratory equipment, and the process would
33 require an immense amount of electricity or power during the extraction, mixing, and curing
34 process (the last step to get the toxin into the white paste). While it is possible for someone to
35 have done this in their home, I cautioned Detective Reyne that the toxin was very likely created in
36 a lab or facility of some kind. The next day, Detective Reyne came back with more things for me
37 to review after executing a search warrant. One was a sample of residue found near a makeshift
38 chemistry lab in the basement of the home, which tested positive for Clartaxin, but it was only
39 trace amounts. I examined the other evidence and noted there was detergent and deodorant that
40 could have been used to create the toxin, although both of those brands are found in nearly every
41 store in Texoma. While the police did not find any of the banned herbicides, I reviewed pictures
42 of the basement laboratory in Exhibit 6. The lab equipment in the basement was a fairly serious
43 small-scale chemistry workbench, not just a few pieces of basic glassware. All of the things you
44 can see in that picture are used for heating, stirring, distilling (or refluxing), separating, and
45 measuring liquids. You can see glass distillation / reflux apparatus, condensers, retort/ring stands
46 with rods and clamps, a magnetic stirrer, large round-bottom flask, and reagent (or storage) bottles.

1 I confirmed to the Detective that the equipment in that picture was precisely what could be used
2 to make Clartaxin, although none of the equipment you see in Exhibit 6 is illegal or unlawful to
3 own, to be clear.

4
5 At the time, I understood Detective Reyne and the police were looking for a teenage boy who lived
6 at the home with the basement laboratory, but I didn't follow any of that process since I'm just the
7 medical examiner. But sometime after Thanksgiving I was asked if I would be able to create
8 Clartaxin using the equipment like they had in the basement laboratory. I confirmed that I could
9 but it would be much safer to do that in an official, secure lab like the one at my office. But I was
10 told that the purpose of doing the experiment inside the home was to confirm its feasibility and
11 determine how noticeable it would've been for anyone in or around the home. I was also given a
12 copy of an E-Mazon order log in Exhibit 9 and asked if anything listed there could be used in the
13 process to create Clartaxin. Some certainly could – like the deodorant, detergent, beakers, reagent
14 bottles, rubbing alcohol, coffee filters, kaolin clay, potassium carbonate, sodium hydroxide,
15 potassium bicarbonate, and diatomaceous earth.

16
17 I agreed to do the experiment and the district attorney worked with the DEA to obtain one of the
18 banned herbicides from France for this experiment. Yes, the same one in Exhibit 7 by number 9.
19 I also used the exact same type of equipment, deodorant, and detergent the police found in the
20 home and various items listed in the E-Mazon order in Exhibit 9. I got started in the basement of
21 the Quartz home on December 3 at 10 a.m. That house is the one on the left side of Exhibit 1.
22 And the basement in that house looked exactly like what you see in Exhibit 6 – except all of the
23 equipment had been taken by the police at that point. I know because I tested all of it. Shockingly,
24 there was no trace of Clartaxin in any of the equipment anywhere. Just that one residue spot.

25
26 The first step in the overall process was to extract the necessary chemicals from the three raw
27 ingredients. I began with the French herbicide obtained by the DEA. There's a specific compound
28 called Quantaprat, which is a highly toxic, fast-acting, and non-selective chemical that is in
29 herbicide use for controlling weeds. To separate the Quantaprat from the herbicide, you need to
30 use a process called adsorption. That is the adhesion of atoms, ions, or molecules from a gas,
31 liquid, or dissolved solid to the outer or inner surface of a material, forming a thin film. Quantaprat
32 is a positive ion in solution, which makes it strongly attracted to negatively charged surfaces, like
33 clay materials. Using a separatory funnel, you can mix the herbicide and an aqueous solution
34 containing clay particles. You can create that by mixing water with kaolin clay, which is sold in
35 powdered form, and the Quantaprat ions will bind to the clay solution. After that, it's just a
36 question of draining the solution and applying heat to evaporate the liquids. The Quantaprat ions
37 will have created a thin layer around the clay which if carefully removed will leave just the
38 Quantaprat. This is the exact process I followed, and it took three hours and six minutes.

39
40 Next was the deodorant. Nearly every commercial deodorant contains synthetic, antimicrobial
41 chemicals to kill germs. Extraction of these is extremely simple. Scrape the deodorant into a glass
42 container and add rubbing alcohol. The alcohol will dissolve the waxy bases to the deodorant and
43 the added organic fragrances. Gently heat the mixture in a hot water bath to melt everything down
44 without boiling off the solvents but remember to stir thoroughly. This is the hardest part and
45 requires the most patience and focus. Then, using another glass container, pour the warm mixture

1 through a coffee filter capturing the solids and leaving the liquid. After that, like the herbicide, just
2 apply a lot of heat to evaporate the water away. This step took me an hour and a half to finish.
3

4 The final part of the extraction process for me was the detergent – but you could have done the
5 process in any order among the three ingredients. However, as I noted before, you have to mix
6 them relatively quickly once you have finished the three extractions, within 12 hours. So, someone
7 wouldn't need to do the three extractions and mixing process one right after the other, but they
8 also couldn't do it over three or four days, if that makes sense. As for my third extraction step,
9 household chemical detergents contain certain human carcinogens that are synthetic, colorless
10 liquids. These carcinogens mix with water, so traditional separation techniques like I employed
11 before would be useless. Instead, I used a really strong, highly soluble salt made from potassium
12 carbonate and sodium hydroxide, two items on the E-Mazon log. Adding that mixture dramatically
13 increased the ionic strength in the detergent, which strips water molecules away to hydrate the ions
14 forcing the carcinogens into their own distinct organic layer. The last step is, you guessed it, heat
15 – with evaporation leaving the carcinogens in a solid, powder form. The third part in the extraction
16 process was complete just before 4:00 p.m.
17

18 I took a ten-minute break to get some fresh air and chat with the cops who were guarding the house
19 outside, for their own protection. Then I began the mixture process. I did allow Detective Reyne
20 in the basement at this time because they needed to know the intensity of the sulfur aromas – they
21 were also a little annoyed they had not smelled any yet. Once combined, it would take about
22 twelve hours under constant heat and occasional mixing to combine and cure. As soon as I
23 combined the three extracted chemicals, there was instantly a rotten egg smell. I opened the
24 window above the table where I was creating the toxin at Reyne's suggestion, which you can see
25 in Exhibit 8, and that greatly helped ventilate the basement and reduce the smell. I worked until 4
26 a.m., and the smell still was very pungent over those twelve hours. I only needed to check on and
27 mix the solution about every couple of hours once the mixing process started; I did not have to
28 stand there the whole time watching.
29

30 I then took a catnap in my car, waking at 6:30 a.m. sharp, just habit. I didn't smell any sulfur
31 aromas when I went back into the basement. I mixed several times over the next twelve hours
32 with continuous low heat, checking on it about every three hours, and the smell returned about
33 noon that day. Worse than before, like rotten eggs died inside of more rotten eggs. I threw up at
34 one point, and I haven't done that since medical school. It was that bad. The mixture had fully
35 cured into a white paste by 9 p.m. that evening and the smell was gone. We took the toxin I created
36 back to the ME office and tested it to confirm – Clartaxin. How anyone missed the smells
37 associated with that process is unbelievable.
38

39 Further affiant sayeth naught.

40 *Logan Cobb, M.D.*

41 LOGAN COBB, M.D.

42
43 **SIGNED AND SWORN TO BEFORE ME** the undersigned authority on this the 15th
44 day of December, 2025.
45

ADDENDUM TO STATEMENT OF LOGAN COBB, M.D.

With additional study since my original statement, I have some more information to confirm. Making Clartaxin the way I did in the experiment, which was consistent with all of the equipment and supplies found in the basement, so it has to be how Grady did it – even though we cannot be certain – would have produced a toxin with full lethality for about two weeks. At that point, it would start to degrade to the point where it was essentially harmless after the fourth week. Based on half-life analysis on the Clartaxin found in Younger’s body, the residue obtained from the Quartz basement, and the toxin I created during the experiment, I am medically certain that the toxin used to kill Younger was created – meaning finally cured; ready to go – nine to fourteen days before Younger’s first exposure to it. I am certain of that range, but there is no test I’m aware of that could get any more precise than that.

As you may have heard, one of Charlie Quartz's neighbors, Maurice Whitacre, was found dead by police. He'd been dead for several weeks, and the police only found him because the guy's family asked for a welfare check. He was a hermit but still always came for family Christmas dinner. I was called to the scene after some officers noticed a weird smell and knocked down the front door. That house is the one on the right side of Exhibit 1, next door to the Quartz property. The body was in late stages of decomposition, but it was clear Whitacre had a heart attack, had fallen, and been there for many weeks. I saw no signs of foul play on the body, and the police found no signs of forced entry (other than their own). Whitacre's family urged me not to do an autopsy. I had the final decision under the law on whether to do an autopsy, but I saw nothing at the scene or on the body that gave me any suspicion the death was anything other than natural causes – with a really late discovery. It was also Christmas Day. I released the body to the family and understand they had their father cremated. The only thing that I remember from that case was finding a silver laser pointer in Mr. Whitacre's pockets when we examined the body on scene. We also found a box full of the same banned French herbicide as the police found in Grady's secret cubby hole. Yes, the same one you see in Exhibit 7. Maybe Whitacre used it for his corn crops.

This concludes my sworn testimony.

Logan Cobb, M.D.

LOGAN COBB, M.D.

SIGNED AND SWORN TO BEFORE ME the undersigned authority at 8:00 a.m. on this
the day of trial.